

Automated System/UI Testing for Mobile Apps using **Detox**

Introduction:

Detox is an automated System/UI Testing tool for Mobile Apps. Conversely, detox is considered as a grey box testing framework which means it contains both white and black box testing aspects [1]. White box testing means it inspects the code of the application. In contrast, black box testing analyses the functionality of the application according to the specific requirements. More importantly, detox is a framework that is used for automated testing for mobile applications which detects UI and code anomalies [2]. Ultimately, this framework allows the application to function in a more consistent manner which will enhance the overall user experience.

Features:

1. Detox is a cross-platform testing framework which can be used for Android and iOS. This is beneficial for the project since the app is being developed as an Android application [2].
2. Detox can test UI features simulating end-to-end user interactions [3].
3. Detox can test for code inconsistencies within the applications [2].
4. Detox can automatically synchronize to the application behaviour [3]. Note: This feature is unconventional compared to other testing frameworks.

Advantages:

1. Fast execution speed when running the automated tests.
2. Many APIs are bundled to test user interaction with applications [3].
3. Testing synchronously indicates after a test is completed it waits for network requests and animations [1]. This allows the test to be suitable and give accurate information about the application.

Disadvantages:

1. Detox has a limited scope implying it can only be tested on react-native applications.
2. According to BrowserStack, detox has a complex setup process such as configuring files and requires prerequisites in order to run detox [4].

Conclusion:

In all, Detox is an incredible framework that can be used in the react-native project for testing the application features and infrastructural code. More importantly, this testing framework serves as a grey box testing framework. In addition, it is extremely efficient when it comes to synchronizing the application behaviour when performing a test. This tool should be used in the project since it can test the application effectively as mentioned above and it is compatible with the application that's currently being developed.

References:

- [1] "End to End Detox Testing Tutorial," BrowserStack. Accessed: Feb. 07, 2025. [Online]. Available: <https://www.browserstack.com/guide/detox-testing-tutorial>
- [2] "Appium vs Detox: Which to Choose in 2023 (Pros & Cons!) | LIZARD.global." Accessed: Feb. 08, 2025. [Online]. Available: <https://lizard.global/blog/appium-vs-detox-which-to-choose-in-2023-pros-and-cons>
- [3] "Detox vs Appium: Which is Best for Your Project?" Accessed: Feb. 08, 2025. [Online]. Available: <https://testsigma.com/blog/detox-vs-appium/>
- [4] "Run your first automated Detox Android UI Tests | BrowserStack Docs." Accessed: Feb. 08, 2025. [Online]. Available: <https://www.browserstack.com/docs/app-automate/detox/getting-started>